195347

Course Name: Probability and Statistics

Course Outcome

CO1- Know the basics of Statistics and apply these techniques in the core course of their study.

CO2- Understand the fundamental concepts of random variables, and their expectations.

CO3- Understand and apply discrete and continuous probability distributions in real-life situations.

CO4- Apply the Central Limit theorem in real life scenarios.

CO5- Understand and model the relationship between two variables.

C06- Perform the application of statistical inference in decision-making situations.

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End Term Examination, Odd Semester 2024-25
B.Tech. (C.S.), II Year, III Semester
Subject Code & Subject Name- BMAS 0108 & Probability and Statistics

Time: 3 Hours **Maximum Marks: 50**

Instruction for students:
Attempt ALL sections.

Section – A

Attempt All Questions. 4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Draw a histogram for the following table: **Class : 2-4 4-8 8-14 14-16 16-20 20-24 24-26** **Frequency : 3 56 144 41 52 32 2**	4	1	R	F
2	Eight coins were tossed together and the number of heads resulting was noted. The operation was repeated 256 times and the frequencies (f) that were obtained for different values of x, the number of heads, are shown in the following table. Calculate median and quartiles. x: **0 1 2 3 4 5 6 7 8** f: **1 9 26 59 72 52 29 7 1**	4	1	R	F

3	From a lot of 10 items containing 3 defectives, a sample of 4 items is drawn randomly. Let the random variable X denote the number of defective items in the sample. Answer the following when the sample is drawn without replacement. a) Find the probability distribution of X. b) Find P (0 < X < 2).	4	2	U	C
4	The average test marks in a particular test is 79. The standard deviation is 5. If the marks are normally distributed, how many students in a class of 200 did not receive marks between 75 and 82? (Given that: $P\ (0 \leq Z \leq 0.8) = 0.2881$ and $P\ (0 \leq Z \leq 0.6) = 0.2881$, where Z is a standard normal variate.)	4	3	U	C
5	A machine produces items, and each item has a 5% chance of being defective. The items are checked one by one. Let X represent the number of inspections until the first defective item is found. What is the probability that the first defective item is found on the 7th inspection? **OR,** In a book of 520 pages, 390 typo-graphical errors occur. Assuming Poisson law for the number of errors per page, find the probability that a random sample of 5 pages will contain no error.	4	3	U	M

Section – B

*Attempt **All** Questions.* 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	A pair of dice is thrown 6 times. If getting a total of 9 is considered a success, what is the probability of getting at least 5 successes? **OR,** Give the probability function of distributions which follow memory-less property.	3	3	A	F

7	Out of the following two regression lines, $40x-18y=320$, $8x-10y=64$, determine which one of these is the regression line of x on y and which one is that of y on x. Also find correlation coefficient between x and y.	3	5	An	C
8	If a random variable X follows a Poisson distribution such that **P(X = 1) = 2 P(X = 2),** Find mean and variance of X. Also find P (X = 0).	3	3	A	P
9	Calculate Karl Pearson's correlation coefficient from the following data: $n=10,\ \sum X=140,\ \sum Y=150,\ \sum (X-10)^2=180,$ $\sum (Y-15)^2=215$ *and* $\sum (X-10)(Y-15)=60.$	3	5	E	P
10	Ten individuals are chosen at random from a population and their heights are found to be **63, 63, 64, 65, 66, 69, 69, 70, 70, 71** inches. Test if the sample belongs to the population whose mean height is 65 inches. [Given $t_{0.05}$ = 2.62 for 9 d.f.]	3	6	C	M

Section – C

*Attempt **All** Questions.* 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
11	Find: (i) the two regression coefficients, (ii) the regression lines, and (iii) the coefficient of correlation for the following data: **Price (X) : 78 89 97 69 59 79 68 61** **Demand (Y) : 125 137 156 112 107 136 123 108**	5	5	E	P

<table>
<tr>
<td>12</td>
<td>A manufacturer who produces medicine bottles finds that 0.1% of the bottles are defective. The bottles are packed in boxes, each containing 500 bottles. A drug manufacturer buys 100 boxes from the producer of bottles. Using Poisson distribution, find how many boxes will contain

(i) no defective bottles,
(ii) at least two defective bottles.

(use $e^{-0.5} = 0.6065$)</td>
<td>5</td>
<td>3</td>
<td>A</td>
<td>P</td>
</tr>
<tr>
<td>13</td>
<td>1600 families were selected at random in a city to test the belief that high income families usually send their children to public schools and low income families often send their children to government schools. The followings results were obtained in the study conducted:

<table>
<tr><th>Income</th><th>Public school</th><th>Government School</th><th>Total</th></tr>
<tr><td>Low</td><td>494</td><td>506</td><td>1000</td></tr>
<tr><td>High</td><td>162</td><td>438</td><td>600</td></tr>
<tr><td>Total</td><td>656</td><td>944</td><td>1600</td></tr>
</table>

Examine by χ^2 test to ascertain if the income of the families and type of schools are independent. (Given 5% value for χ^2 for 1 d. f. is 3.84)</td>
<td>5</td>
<td>6</td>
<td>An</td>
<td>C</td>
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